

Complement C1q drives microglia-dependent synaptic loss and cognitive impairments in a mouse model of lipopolysaccharide-induced neuroinflammation

Xinmiao Wu^a, Yuzhu Gao^a, Cuina Shi^a, Jianhua Tong^a, Daqing Ma^b, Jinchun Shen^{c,***}, Jianjun Yang^{d,**}, Muhuo Ji^{a,*}

^a Department of Anesthesiology, The Second Affiliated Hospital of Nanjing Medical University, Nanjing, China

^b Division of Anaesthetics, Pain Medicine and Intensive Care, Department of Surgery and Cancer, Faculty of Medicine, Imperial College London, Chelsea and Westminster Hospital, London, UK

^c Department of Anesthesiology, Jinling Hospital, Medical School of Nanjing University, Nanjing, China

^d Department of Anesthesiology, Pain and Perioperative Medicine, The First Affiliated Hospital of Zhengzhou University, Zhengzhou, China

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ABSTRACT

Activated microglia and subsequent release of pro-inflammatory cytokines result in neuroinflammatory status which further damage neurological function including cognitive impairments in various neurological conditions. However, the underlying molecular mechanisms during these pathological processing remain unknown. In the current study, mice received intraperitoneal administrations of LPS (0.5 mg/kg, daily, *Escherichia coli* O55:B5) for seven consecutive days and their different cohorts were used for behavioral assessment with open field, Y maze, and novel object recognition test or for electrophysiology recordings of mEPSC, LFP or LTP in *in vivo* or *ex vivo* preparation. The hippocampus from some cohorts were harvested for immunostaining or Western blotting of c1q, Iba-1, CD68, PSD95 and dendritic spine density or for transcriptome and proteomics analysis. Repeated LPS injections induced an up-regulation of complement system protein c1q and distinct microglial phenotype with an enrichment of the complement-phagosome pathway. Microglial synaptic engulfment and profound synaptic loss were found. These pathological changes were accompanied with the significantly decreased excitatory synaptic transmission, disturbed theta oscillations, impaired hippocampal long-term potentiation, and cognitive impairments. Notably, neutralization of c1q signaling robustly prevented these changes. Collectively, our data provide evidence that activated microglia and complement cascade c1q signaling in the hippocampus may account for synaptic loss and cognitive impairments in a mouse model of neuroinflammation induced by repeated LPS injections. Our work implicates that complement system may be a therapeutic target for developing therapies to prevent or treat cognitive disorders related to neuroinflammation or other disease conditions including neurodegenerative disease *per se*.

Author statement

Xin-miao Wu: Methodology, Formal analysis, and Writing-Original Draft, **Yu-zhu Gao:** Methodology, Writing-Review & Editing and Funding acquisition, **Cui-na Shi:** Methodology and Writing-Review & Editing, **Jian-hua Tong:** Methodology and Writing-Review & Editing, **Da-qing Ma:** Writing-Review & Editing, **Jin-chun Shen:** Conceptualization and Editing, **Jian-jun Yang:** Conceptualization and Editing, **Mu-**

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1. Introduction

Over activated microglia driven ill-state of neuroinflammation results in cognitive impairments in many neurodegenerative diseases and psychiatric disorders, including Alzheimer's disease (AD), postoperative

* Corresponding author. Department of Anesthesiology, The Second Affiliated Hospital, Nanjing Medical University, Nanjing, China.

** Corresponding author.

*** Corresponding author.

E-mail addresses: yyshen0203@163.com (J. Shen), yjyangji@126.com (J. Yang), jimuhuo2009@sina.com (M. Ji).

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cognitive impairment, sepsis-associated encephalopathy, delirium, and depression (Cheng et al., 2022; Ji et al., 2015, 2021; Qiu et al., 2016; Taylor et al., 2022; Teunissen et al., 2022). Systemic inflammation can reach to any vital organs including the central nervous system (CNS) to activate microglia which, if persist, initiate neurodegenerative disease development (Odoj et al., 2021; Süß et al., 2020). Indeed, neuroinflammation with excess microglial activation and synaptic impairment are early and key hallmarks underlying cognitive impairments of many neurodegenerative diseases (Puntambekar et al., 2022; Qu and Li, 2020; Wu et al., 2020). Furthermore, previous studies have demonstrated that a peripheral inflammatory challenge subsequently caused excess microglia activation and robust hippocampal-dependent memory impairments (Izumi et al., 2021; Tyrtysyna et al., 2022), yet the precise mechanism remains unclear. Therefore, a better understanding of the mechanisms by which how activated microglia damages brain functions may help to develop therapies to prevent or treat cognitive disorders related to neuroinflammation or other types of neurodegeneration diseases.

In an animal model of depression, lipopolysaccharide (LPS) induced microglia activation and reduced the number of dendritic spines and brain-derived neurotrophic factor expression, pre-synaptic protein synaptophysin and post-synaptic density protein 95 (PSD95) (Li et al., 2021). Systemic inflammation is also associated with a reduction in synaptopodin expression in the hippocampus (Strehl et al., 2014). Excitatory synapse damage in the hippocampus in association with activated microglia was observed in a mouse model of multiple sclerosis (Bellizzi et al., 2016) and a septic model induced by cecal ligation and puncture (Basak et al., 2021). Clinically, synaptic loss together with reactive gliosis were also considered to be a common pathological feature of neurodegenerative diseases (Harada et al., 2022; Pereira et al., 2021; Zhao et al., 2020). However, it remains unclear how activated microglia ultimately caused synaptic injury and cognitive impairments.

Microglial regulation of activity-dependent plasticity and synaptic remodeling in neuronal networks has been the subject of recent research (Du et al., 2022; Guedes et al., 2022). During brain development, microglia are critical for maintaining neuronal homeostasis and optimal neuronal plasticity (Paolicelli et al., 2011). Under pathological state, microglia responded to CNS injury by becoming reactive and increasing phagocytic activity, leading to synaptic pruning by engulfing and removing synapses (Dayananda et al., 2022; Hong et al., 2016). One key mechanism of those is the classical complement cascade dependent phagocytic signaling (Hong et al., 2016). Indeed, classical complement cascade-mediated phagocytic signaling is aberrantly up-regulated in mouse models of AD, frontotemporal dementia, and West Nile virus infection, leading to synapse loss (Hong et al., 2016; Vasek et al., 2016). However, mice deficient with CR3, C3 or C1q expression in microglia had a significant decreased ability to engulf and remove synapses (Bialas and Stevens, 2013; Schafer et al., 2012). However, it remains unclear if and how these complements and/or microglia modulate synaptic connectivity in response to neuroinflammation derived from systemic inflammation. The present study was designed to investigate that activated microglia associated with complement cascade signaling in the hippocampus may account for synaptic loss and cognitive impairments in a mouse model of neuroinflammation induced by repeated LPS injections.

2. Materials and methods

2.1. Animals

All the animal experiments were approved by the Institutional Animal Care and Use Committee of Nanjing Medical University, Jiangsu, China. Male young adult C57BL/6 mice with 12-week old were obtained from Model Animal Research Center of Nanjing University, Jiangsu, China. Mice were housed 4–5 per cage and maintained at 22–25 °C on a 12 h light/dark cycle. Food and water was obtained *ad libitum*. Mice

were randomly allocated to different experimental groups. All animal experiments were in accordance with the principals of 3Rs (refinement, reduction, and replacement), thus minimizing the number of mice used and maintaining statistical power.

2.2. Animal model of neuroinflammation

Mice received intraperitoneal administrations of LPS (0.5 mg/kg, daily, *Escherichia coli* O55:B5, L2280, sigma) daily between 9 and 11 a. m. for seven consecutive days as reported previously (Yang et al., 2018). Control mice received the same volume of normal saline (0.9%) for seven consecutive days. The flow chart of study procedures is shown in Fig. S1.

2.3. Transcriptome sequencing analysis

Mice were anesthetized by 1.25% tribromoethanol (0.3 ml/10 g) and killed. Their hippocampi were surgically removed for transcriptome sequencing analysis. Total RNA from the sample tissues was immediately extracted with the TIANamp RNA Isolation Kit (TIANGEN) according to the manufacturer's instructions. Then a total of 1 µg RNA per sample was used for the RNA sample preparations and mRNA sequencing library generation. The mRNA was purified by TruSeq RNA Sample Prep Kit (Illumina) according to the manufacturer's protocol. Poly-A mRNA selection, enzymatic fragmentation, first and second strand cDNA synthesis, end-repair, 3'adenylation, adapter ligation, and PCR amplification were included in these processes. The raw RNA reads were processed using in-house perl scripts to discard the reads containing ploy-N, adaptor sequences and low-quality bases. All of the downstream analyses were based on high quality clean data. Clean data were subjected to the reference genome sequence using TopHat. Gene expression values were normalized as fragments per kilobase of transcript per million (FPKM). The DESeq R package was used to identify differentially expressed genes in all the transcriptomic data. The resulting *P* values were adjusted using Fisher's exact test for controlling the false discovery rate. Genes were identified as having significantly differential expression with a threshold of an adjusted *P* value < 0.05 and $|\log_2(\text{FC})| > 1.2$ (Yang et al., 2018).

2.4. Proteomics analysis

Treated cohort mice were anesthetized by 1.25% tribromoethanol (0.3 ml/10 g) and the hippocampi were removed on ice immediately. The hippocampi were homogenized with lysis buffer containing protease inhibitor phenylmethanesulfonyl fluoride and centrifuged at 16,000×g for 30 min. Total supernatant protein concentrations were determined by BCA assay and then subjected to sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). Subsequently, 100 µg of protein from each sample were processed according to the manufacturer's instruction for the TMT reagent kit. Next, the protein were mixed well with 1% SDS (1 µl), and 2 µl reducing reagent was added and incubated at 60 °C for 1 h. After which, cysteine blocking reagent was added and incubated for 10 min under room temperature, and then digested overnight at 37 °C by trypsin with an enzyme/protein ratio of 1:50 and then labeled with TMT reagent. The labeled samples were reconstituted with UPLC loading buffer and separated by high-pH liquid with a reversed-phase C18 column. A total of 20 fractions were collected and then combined into 10 fractions. Subsequently, these fractions were dissolved in mass spectrometry loading buffer for second-dimensional analysis by liquid phase tandem mass spectrometry. The MS raw data for each sample were searched using the MASCOT engine (Matrix Science, London, UK; version 2.2) embedded into Proteomics Discoverer 1.4 software with the Uniprot Mouse database for identification and quantitation analysis. The final proteins that were deemed to be differentially expressed were filtered as a *P* value < 0.05 and 1.2-fold changes (>1.20 or <0.83) relative to the control group

(Noriega-Rivera et al., 2022). Following annotation steps, the studied genes were blasted against the online Kyoto Encyclopedia of Genes and Genomes (KEGG) (<http://geneontology.org/>) to retrieve their KEGG orthology identifications and were subsequently mapped to pathways in KEGG.

2.5. Western blot analysis

After behavioral tests (see below), the whole hippocampal samples were harvested for further process described as above to obtain protein samples. All normalized samples were separated by 10% or 12% SDS-PAGE, and then transferred to PVDF membranes, blocked by 3% skimmed milk, and incubated with primary antibodies overnight at 4 °C, including C1q (ab11861, Abcam), GAPDH (60004-1-Ig, Proteintech), IL-1 β (16806-1-AP, Proteintech), TNF- α (60291-1-Ig, Proteintech) and β -actin (66009-1-Ig, Proteintech). The PVDF membranes were washed with TBST buffer three times and then incubated with secondary antibodies for 2 h at room temperature. The PVDF membranes were then visualized with enhanced chemiluminescence western blotting detection reagents. The protein bands were scanned with ChemiDOCTM MP Imaging System (Bio-Rad), and the gray value of protein band intensity was quantified using ImageJ software. C1q, IL-1 β , and TNF- α levels were normalized to β -actin and GAPDH.

2.6. Stereotaxic injection of C1q-neutralization antibody

Mice were anesthetized with 1.25% tribromoethanol (0.3 ml/10 g) and placed in a stereotaxic frame. A microwavable heating pad was used to maintain body temperature and sterile ointment was applied to each eye to avoid corneal drying. The target site CA1 was defined using stereotaxic coordinates (AP, -2.0 mm; ML, $\pm$ 1.5 mm; DV, -1.5 mm). Solution of C1q-neutralization antibody (0.15 μ g, ab11861, Abcam) (Figueiredo et al., 2019) was micro-injected with a glass pipette at a flow rate of 0.15 μ l/min.

2.7. Behavioral tests

2.7.1. Open field test

The open field test was performed as we previously described (Wu et al., 2022). Briefly, mice were introduced into the center of a white experimental box (50 $\times$ 50 $\times$ 50 cm) and allowed to explore the area freely for 5 min. The total distance traveled and the time spent in center area were automatically recorded with a video tracking system (RWD Life Science Inc). At the end of each trial, the arena was cleaned with 75% alcohol to eliminate olfactory cues. In all the behavioral tests, the investigator was blinded to animal grouping during the experiment and quantitative analyses.

2.7.2. Y-maze test

The Y-maze test procedure was conducted as previously described (Wu et al., 2022). The Y-maze consisted of three arms and each arm was labeled as A, B, C. Mice were placed in the center of the maze and explored freely throughout three arms for 8 min. The spontaneous alternation performance were recorded. The alternation was defined as successive consecutive entries to three different arms. The percentage of spontaneous alternation is the number of triads containing entries into all three arms divided by the maximum possible alternations (the total number of arms entered minus 2) $\times$ 100.

2.7.3. Novel object recognition test

The test was carried out in a white experimental box measuring 50 $\times$ 50 $\times$ 50 cm as previously described (Zhang et al., 2021). To minimize the stress of bright lighting, diffuse and low lighting was adopted. Test objects were mouse-sized and different enough to be easily discriminated. During behavioral sessions, objects were fixed to the box using tape to prevent displacement caused by exploratory activity of the

animals. Before training, each animal was submitted to a 5 min-long habituation session, in which it was allowed to freely explore the arena. During the 10-min training session, mice were placed at the center of the arena in the presence of two identical objects. The arena and objects were cleaned thoroughly with 75% ethanol between trials to eliminate olfactory cues. Twenty-four hours after training, animals were again placed in the arena for the test session, with one of the objects used in the training session replaced by a new object. Exploratory behavior was defined as sniffing and touching the object. The discrimination index was defined as [(novel object exploration time)/(total exploration time of both objects) $\times$ 100]. All trials were recorded and analyzed by the computerized tracking system (RWD Life Science Inc).

2.8. Golgi staining

Golgi silver impregnation was done using a commercially available kit (FD Rapid GolgiStain Kit, FD NeuroTechnologies) as we previously described (Wu et al., 2022). Mice were anesthetized with tribromoethanol (0.3 ml/10 g), and their brains were harvested as quick as possible. Coronal sections (150 μ m) were cut using a vibratome (model VT1000S, Leica), and then mounted on gelatin-coated slides and developed following the protocol in the FD Rapid GolgiStain Kit. CA1 pyramidal neurons from the dorsal hippocampus were imaged using microscopy (Olympus, Tokyo, Japan) with a 20 $\times$ objective and 100 $\times$ oil objective. Neuronal morphology was analyzed with plugin NeuronJ and sholl analyses in Image J software. Dendritic spines were counted using the Cell Counter plugin in Image J software. For analysis, three CA1 pyramidal neurons and their spines were randomly chosen from each animal. Spine density was calculated as the number of spines divided by the length of the part of the dendrite from which the spines were counted.

2.9. Whole-cell patch-clamp recordings

Whole-cell patch-clamp recordings were performed as we previously described (Ji et al., 2020a). Briefly, brain slices harvested from mice were placed in a recording chamber and perfused with artificial cerebrospinal fluid, which with bubbled with 95% O₂/5% CO₂. Whole-cell recordings were performed on pyramidal neurons in CA1 visualized with infrared optics using an upright microscope equipped with a $\times$ 40 water-immersion lens (Olympus BX51W1, Japan) and an infrared-sensitive CCD camera. The patch pipette (input resistance: 3–6 M Ω) solution contained with 125 potassium D-gluconate mM, 8 NaCl, 0.2 EGTA mM, 10 HEPES mM, 2 Mg-ATP mM, 0.3 NaGTP mM. Series resistances were usually 15–30 M Ω upon break-in and were compensated by 70%. Pyramidal neurons with stable series resistance (20% changes throughout the recording) were used for analysis. Data were collected by an Axon patch 700B amplifier, low-pass filtered at 2 kHz, digitally sampled at 10 kHz and analyzed with Clampfit software (Molecular Devices). To measure the intrinsic membrane properties of neurons, spiking patterns were recorded in the current-clamp configuration by injecting a series of current pulses (400 ms duration, 60–240 pA intensity with an increment of 60 pA). To isolate miniature EPSCs (mEPSCs), tetrodotoxin 1 and bicuculline 10 μ M were added to the bath solution. mEPSCs were analyzed using the Mini Analysis Program (Version 6.0.3, Synaptosoft) and all events were detected above a threshold of 5 pA.

2.10. Immunohistochemistry

Mice were deeply anesthetized by 1.25% tribromoethanol (0.3 ml/10 g) and were transcardially perfused with PBS followed by 4% paraformaldehyde (PFA) solution. Brains were removed and post-fixed overnight in 4% PFA and dehydrated in 20% and 30% sucrose at 4 °C overnight and then embedded in O.C.T. Then brains were cut coronally into 30- μ m-thick sections by freezing microtome. The sections were

blocked with PBS containing 10% normal goat serum for 1 h at room temperature. Next the sections were incubated with primary antibodies overnight at 4 °C, including rabbit anti-Iba1 (019–19741, Wako, 1:500), rat anti-CD68 (ab53444, Abcam, 1:500), rat anti-C1q (ab11861, Abcam, 1:500), mouse anti-PSD95 (MA1-046, thermofisher, 1:500), rabbit anti-PSD95 (PA5-85749, thermofisher, 1:500) and mouse anti-c-fos (sc-166940, santa cruz, 1 : 100). Sections were washed with PBS three times and incubated with goat anti-mouse IgG-FITC (115095146, Jackson ImmunoResearch, 1:400), goat anti-rabbit IgG-Cy3 (115165045, Jackson ImmunoResearch, 1:400), goat anti-rat IgG-FITC (112095003, Jackson ImmunoResearch, 1:400) for 2 h at room temperature. Then sections were then washed with PBS three times, which was followed by incubation with 4,6-diamidino-2-phenylindole (DAPI) for 10 min at room temperature for nuclear staining. Four slices randomly picked from per mouse were imaged and quantified for five to six mice in each group. Fluorescent images were captured by a confocal microscope (Olympus FV3000). For quantification of immunohistochemistry analyses, 20× objective was used for the analysis of cell counts of IBA1⁺ cells and IBA1⁺ colocalization With CD68. For the three-dimensional reconstruction, 40× objective with a zoom value of 3 was used. For the fluorescence quantification of C1q and PSD95, 40 × objective with a zoom value of 4 was used.

As previously reported (Cao et al., 2021), the analysis of microglia morphology was assessed by image J. Briefly, z-stack of images (0.56 µm thick z stacks) was taken by confocal microscope. Then, images were converted to 8-bit and enhanced the contrast by using the function Unsharp Mask (a pixel radius of 3 and mask weight of 0.6). Next, the background noise of image was reduced by using the function Despeckle and Remove Outliers. Then the image was adjusted threshold to binary image and measured. The value of solidity is the ratio of cell body and process area of microglia, which reflects microglia activation. For the analysis of microglia phagocytic function, the overlay of Iba1 and PSD95 puncta was assessed by image J and IMARIS. PSD95⁺ puncta was reconstructed using the IMARIS Spots function. We used the IMARIS MATLAB-based plugin (Split into Surface Objects) to assess the number of PSD95⁺ puncta entirely within the microglial surface. Normalized positive area values were calculated from the positive area value divided by the average positive area value of control group.

2.11. Local field potentials (LFPs) recording and data analysis

The procedure of *in vivo* electrophysiology was performed as we previously described (Ji et al., 2020b). The microelectrode implanting surgery was conducted before LPS injections. Briefly, small holes were drilled into the skull at the position of the recording electrodes after deep anesthesia in the mice. An eight-channel microelectrode (Kd-micro-008, KedouBC) were then implanted through burr holes targeting the CA1 (AP, −2.0 mm; ML, ±1.5 mm; DV, −1.5 mm). The electrode was slowly advanced into the targeting position using a custom-made stereotaxic system. Then a thin layer of adhesive cement was applied to the skull and allowed to dry before dental acrylic was used to cover the skull cap. After repeated LPS injections, the task related LFPs were recorded in the Y maze to investigate changes in neural oscillations in CA1 as we previously described (Cao et al., 2021). Mice were placed in Y maze to record task related LFPs. The timing of behavioral events (Time = 100 ms) indicates the moment when the mice was at the choice point. The signals were transmitted into a digital preamplifier (Plexon Inc., Dallas, TX) and digitized at a sampling rate of 40 kHz. According to the frequency, the signals were separated to delta (0.5–3 Hz), theta (4–7 Hz), alpha (8–13 Hz), beta (14–29 Hz), and gamma oscillations (30–80 Hz). The signals were filtered with a band-pass of 0.3–300 Hz, further amplified and digitized at 2 kHz. For LFP analysis, the wide band signals were down-sampled to 1250 Hz. All data analysis was performed using the software NeuroExplorer 5 (Plexon Inc., Dallas, TX).

2.12. Statistical analysis

Statistical analyses were analyzed by the GraphPad Prism version 8.0. All data were tested for normality with the Shapiro-Wilk normality test and then presented as mean ± S.E.M. Differences between two groups were analyzed with independent-sample *t*-test, whereas differences among multiple means were assessed by one-way or two-way analysis of variance (ANOVA) followed by a Turkey test as appropriate. A significant difference was considered as *P* less than 0.05.

3. Results

3.1. Repeated LPS injections induced increase in C1q expression in the hippocampus

First, we assessed the changes of TNF-α and IL-1β expressions after LPS administration by Western blot. As shown in Figs. S2A–D, up-regulation of TNF-α [$F_{(4, 15)} = 3.364$, $P = 0.0373$; Ctrl vs. 3d: $P = 0.0198$, Ctrl vs. 5d: $P = 0.3045$, Ctrl vs. 7d: $P = 0.0257$, Figs. S2A and C] and IL-1β [$F_{(4, 15)} = 6.000$, $P = 0.0043$; Ctrl vs. 3d: $P = 0.0279$, Ctrl vs. 5d: $P = 0.0203$, Ctrl vs. 7d: $P = 0.0138$, Figs. S2B and D] persisted on day 7 after LPS administration when compared with control group, suggesting our protocol did not induce significant LPS tolerance. To investigate the potential mechanisms underlying LPS-induced cognitive impairments, we conducted transcriptome sequencing and proteomics analysis. As shown in Fig. 1A–B, the significant changes in transcription and protein levels between the control and LPS groups are presented in heat maps. Both transcriptome sequencing and proteomics analysis showed an increased level of C1q expression in the hippocampi after repeated LPS injections. To understand the biological significance of the identified differential proteins found in the LPS group, we performed KEGG pathway analysis, which suggested that the complement cascade pathway was one of the pathways significantly affected by repeated LPS injections [Fig. 1C]. To further validate these, the change of C1q expression in the hippocampi was further validated with western blotting analysis. Consistently, the protein levels of C1q were significantly increased in the LPS group compared with the control group [$t = 5.002$, $P = 0.0022$, Fig. 1D–E].

3.2. C1q-neutralization antibody applied in the hippocampal CA1 improved LPS-induced cognitive impairments

We examined whether C1q-neutralization antibody could block the increased level of C1q. Western blot showed that C1q-neutralization antibody down-regulated C1q expression after LPS administration [Interaction: $F_{(1, 20)} = 5.273$, $P = 0.0326$; LPS, $F_{(1, 20)} = 21.28$, $P = 0.0002$; Rescue, $F_{(1, 20)} = 2.789$, $P = 0.1105$, Figs. S3A–B]. Next, we investigated whether blockage of C1q by specific antibody may improve LPS-induced cognitive impairments. There was no difference in total distance and time in the center of the open field test among four groups [Interaction: $F_{(1, 56)} = 0.1297$, $P = 0.7201$; LPS, $F_{(1, 56)} = 0.08711$, $P = 0.7690$; Rescue, $F_{(1, 56)} = 0.1782$, $P = 0.6746$, Fig. 2A; Interaction: $F_{(1, 56)} = 0.007935$, $P = 0.9293$; LPS, $F_{(1, 56)} = 0.1824$, $P = 0.6709$; Rescue, $F_{(1, 56)} = 0.4218$, $P = 0.5187$, Fig. 2B], which excluded the possible effects of locomotor activity on behavioral results. In the Y-maze test, mice in the LPS + PBS group showed lesser spontaneous alternation than those in the control + PBS group, which was reversed by C1q-neutralization antibody [Interaction: $F_{(1, 56)} = 4.248$, $P = 0.0440$; LPS, $F_{(1, 56)} = 6.116$, $P = 0.0165$; Rescue, $F_{(1, 56)} = 3.087$, $P = 0.0844$, Fig. 2C; Interaction: $F_{(1, 56)} = 0.05780$, $P = 0.8109$; LPS, $F_{(1, 56)} = 1.576$, $P = 0.2145$; Rescue, $F_{(1, 56)} = 0.0007136$, $P = 0.9788$, Fig. 2D]. In the novel object recognition test, mice in the LPS + PBS group showed decreased discrimination index than those in the control + PBS group, which was reversed by C1q-neutralization antibody [Object1 vs Object2: control + PBS: $t = 0.1202$, $P = 0.9063$; control + anti-C1q: $t = 0.1980$, $P = 0.8464$; LPS + PBS: $t = 0.5802$, $P = 0.5725$; LPS + anti-C1q: $t = 1.191$, $P =$

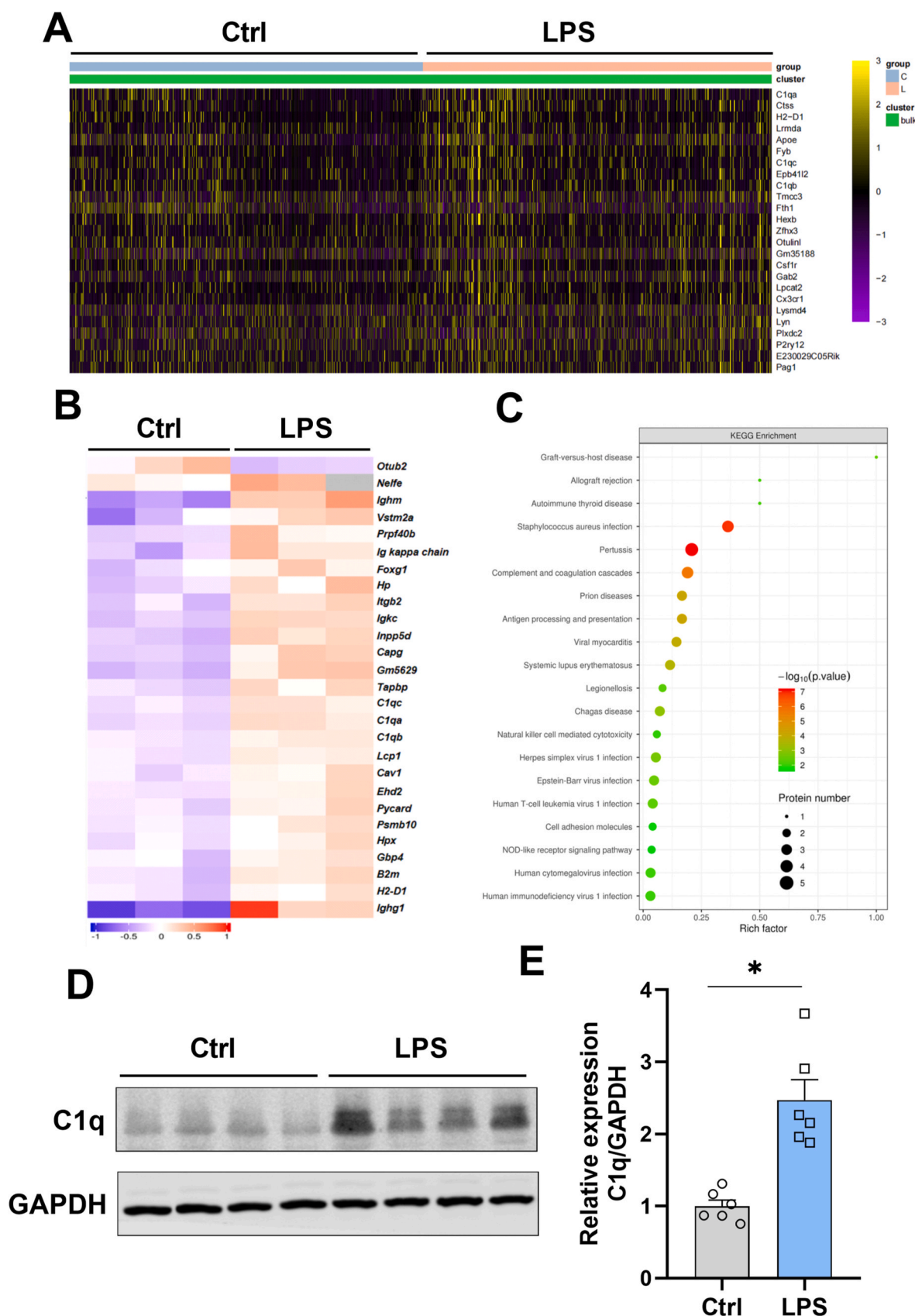


Fig. 1. Repeated LPS injections induced increase in C1q expression in the hippocampi. (A) Hierarchical clustering heat maps of changes in the transcription level between Ctrl and LPS groups. (B) Hierarchical clustering heat maps of changes in the protein level between control and LPS groups. (C) KEGG pathway enrichment of differentially expressed proteins. (D) Representative Western blot of C1q in the hippocampi. (E) Quantitative analysis of C1q level in the hippocampi. Data are presented as the mean $\pm$ SEM ($n = 3-6$), * $P < 0.05$.

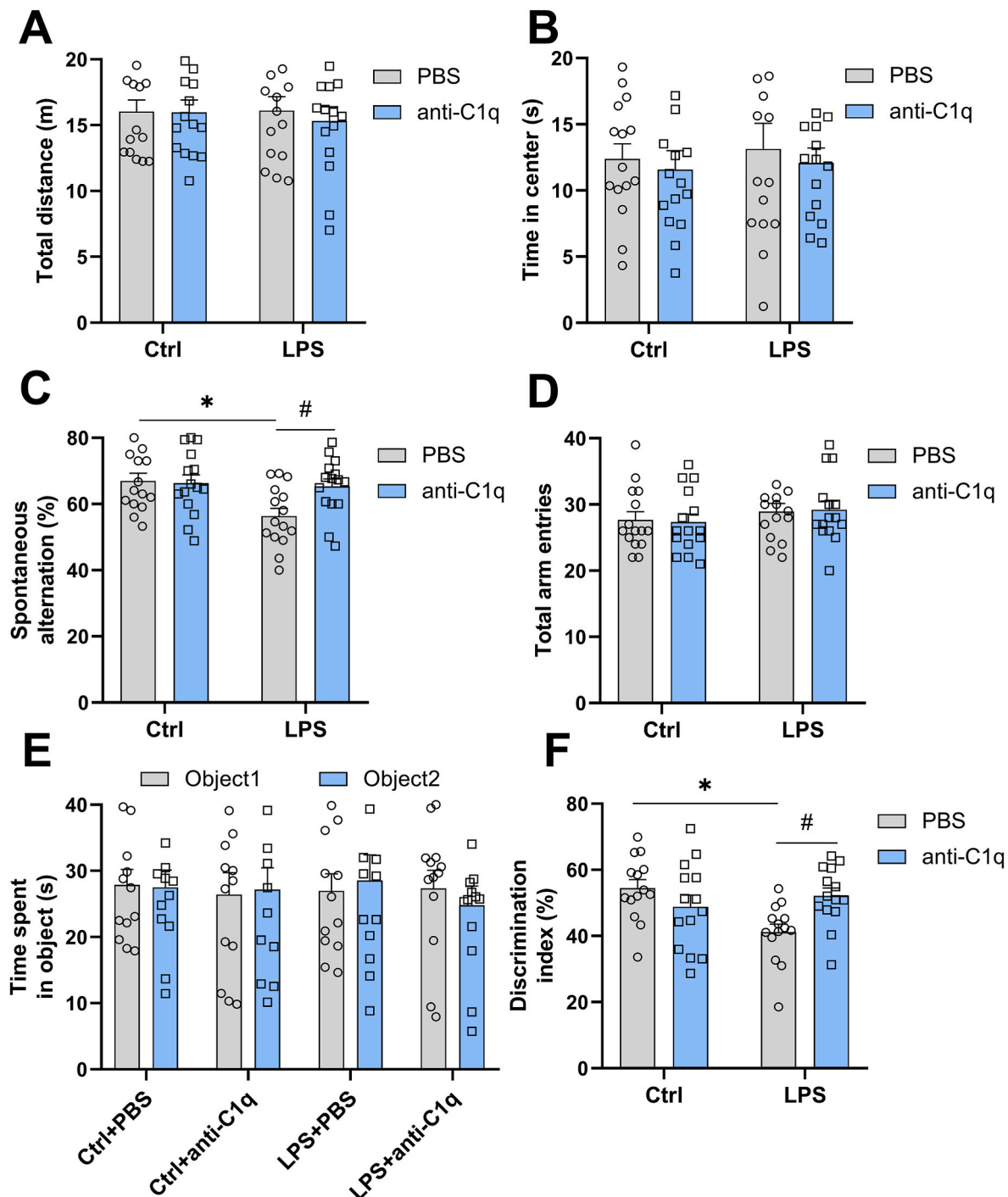


Fig. 2. C1q-neutralization antibody applied in the hippocampal CA1 improved LPS-induced cognitive impairments. (A–B) Performance of open field test among these groups. (C–D) Performance of Y maze test among these groups. (E–F) Performance of novel object recognition test among these groups. Data are presented as the mean $\pm$ SEM ($n = 13$ – 15), * $P < 0.05$, # $P < 0.05$.

0.2567, Fig. 2E; Interaction: $F_{(1, 52)} = 9.011$, $P = 0.0041$; LPS, $F_{(1, 52)} = 3.306$, $P = 0.0748$; Rescue, $F_{(1, 52)} = 0.9454$, $P = 0.3354$, Fig. 2F]. These data suggested that C1q up-regulation played a significant role in working memory impairment after repeated LPS injections.

3.3. C1q-neutralization antibody attenuated LPS-induced dendritic spine loss in the hippocampal CA1

Hippocampal synapse loss is one of the main pathological changes that are responsible for cognitive impairments in many

neurodegenerative diseases. Synapse loss may be due to C1q mediating synapse pruning. Golgi staining showed that no significant differences in total dendritic length and intersection numbers were observed among the 4 groups [Interaction: $F_{(1, 36)} = 1.230$, $P = 0.2747$; LPS, $F_{(1, 36)} = 0.03287$, $P = 0.8572$; Rescue, $F_{(1, 36)} = 0.4693$, $P = 0.4977$, Fig. 3A–C]. The dendritic spine density was significantly reduced in the LPS + PBS group, which was prevented by C1q-neutralization antibody [Interaction: $F_{(1, 39)} = 1.061$, $P = 0.3093$; LPS, $F_{(1, 39)} = 28.89$, $P < 0.0001$; Rescue, $F_{(1, 39)} = 10.23$, $P = 0.0027$, Fig. 3D–E].

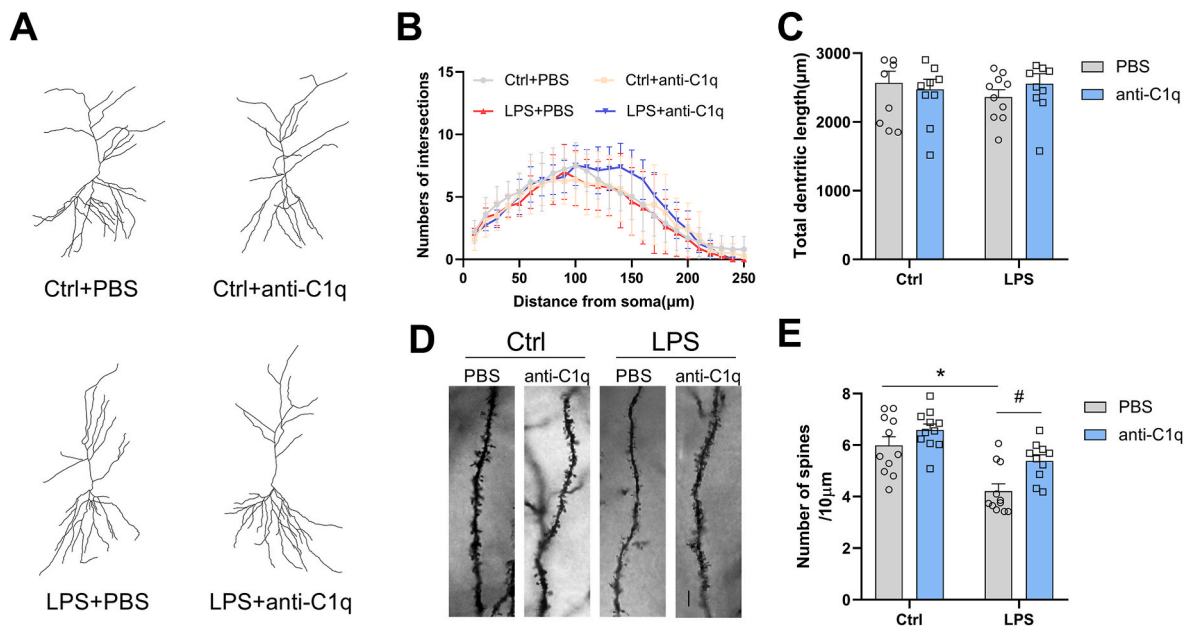


Fig. 3. C1q-neutralization antibody attenuated LPS-induced dendritic spine loss in the hippocampal CA1. (A) Images of tracings of hippocampal neurons. (B) Quantification of dendritic intersections of hippocampal neurons. (C) Quantification of the total dendritic length of hippocampal neurons. (D) Representative dendritic spine density of hippocampal neurons. (E) Quantification of the dendritic spine density of hippocampal neurons. Scale bar = 10 μm. Data are presented as the mean ± SEM (n = 5), * $P < 0.05$, # $P < 0.05$.

3.4. C1q-neutralization antibody attenuated LPS-induced decrease in the membrane excitability and excitatory synaptic transmission in the hippocampal CA1

To determine whether C1q is associated with synaptic transmission dysfunction, we next performed whole-cell patch-clamp recording to investigate the effect of LPS on the excitability of pyramidal neurons. To assess the functional consequences of LPS on synaptic transmission, the basic synaptic transmission of pyramidal neurons in CA1 was determined with the amplitude and frequency of spontaneous miniature EPSCs (mEPSCs) [Fig. 4A]. The mean amplitude of mEPSCs was unaffected among the three groups [control vs. LPS: $P = 0.4520$; LPS vs. LPS + anti-C1q: $P = 0.6206$, Fig. 4B–C], while the mean frequency was significantly decreased in the LPS group [control vs. LPS: $P = 0.0108$; LPS vs. LPS + anti-C1q: $P = 0.0445$, Fig. 4D–E]. In addition, in response to a series of suprathreshold depolarizing current injections (an increment of 60 pA), the number of action potentials recorded was also significantly decreased in the LPS group than that of the control group, which was reversed by C1q-neutralization antibody administration [240 pA: control vs. LPS: $P < 0.0001$; LPS vs. LPS + anti-C1q: $P = 0.0428$, Fig. 4F–G].

3.5. C1q-neutralization antibody attenuated LPS-induced microglia activation in the hippocampal CA1

Given activated microglia is a hall mark of neuroinflammation, we examined LPS-induced microglia activation [Fig. 5A–B]. Compared with the control group, the number of IBA1⁺ cells was increased in the CA1 in the LPS group, which was reversed by C1q-neutralization antibody [Interaction: $F_{(1, 16)} = 12.19$, $P = 0.0030$; LPS, $F_{(1, 16)} = 25.83$, $P = 0.0001$; Rescue, $F_{(1, 16)} = 0.7051$, $P = 0.4135$, Fig. 5C]. We subsequently analyzed the morphological changes of microglial cells, which is known to be feature of microglia activation status. The solidity of microglia in the LPS group was significantly increased [Interaction: $F_{(1, 16)} = 16.44$, $P = 0.0009$; LPS, $F_{(1, 16)} = 19.80$, $P = 0.0004$; Rescue, $F_{(1, 16)} = 2.577$, $P = 0.1280$, Fig. 5D]. Their morphological changes were prevented by C1q-neutralization antibody.

3.6. C1q-neutralization antibody attenuated LPS-induced increase in microglia phagocytosis in the hippocampal CA1

Considering microglia express complement receptors and mediate synaptic pruning in the developing brain, we wonder whether an increased level of C1q might induce microglia excessive synaptic pruning [Fig. 6A–B]. Microglia in the LPS-treated mice had an increased lysosomal protein CD68 of phagocytosis [Interaction: $F_{(1, 16)} = 6.657$, $P = 0.0201$; LPS, $F_{(1, 16)} = 12.04$, $P = 0.0032$; Rescue, $F_{(1, 16)} = 2.865$, $P = 0.1099$, Fig. 6C]. In addition, microglia in the LPS group engulfed more PSD95 puncta than those in the control group, while the C1q-neutralization antibody reversed the excessive phagocytosis [Interaction: $F_{(1, 16)} = 7.624$, $P = 0.0139$; LPS, $F_{(1, 16)} = 8.053$, $P = 0.0119$; Rescue, $F_{(1, 16)} = 2.793$, $P = 0.1141$, Fig. 6D]. These results suggested that the up-regulation of C1q might result in excessive phagocytosis of microglia.

3.7. C1q-neutralization antibody attenuated C1q-mediated microglial recognition and pruning of synapses in the hippocampal CA1

To further investigate the mechanism underlying microglial excessive synaptic pruning induced by repeated LPS injections, C1q was co-labeled with PSD95 [Fig. 7A]. Compared with the control group, the PSD95 puncta was significantly decreased in the CA1 in the LPS group, which was reversed by C1q-neutralization antibody [Interaction: $F_{(1, 16)} = 5.367$, $P = 0.0341$; LPS, $F_{(1, 16)} = 14.38$, $P = 0.0016$; Rescue, $F_{(1, 16)} = 3.050$, $P = 0.0999$, Fig. 7B]. Consistently, mice in the LPS group had an increased level of C1q in the CA1 compared with the control group [Interaction: $F_{(1, 16)} = 3.608$, $P = 0.0757$; LPS, $F_{(1, 16)} = 27.03$, $P < 0.0001$; Rescue, $F_{(1, 16)} = 6.127$, $P = 0.0249$, Fig. 7C]. We further compared the positive area ratio of PSD95 and C1q to reflect the relationship between PSD95 and C1q. Compared with the control group, the positive area ratio of PSD95 and C1q was significantly decreased in the CA1 in the LPS group, which was reversed by C1q-neutralization antibody [Interaction: $F_{(1, 16)} = 5.795$, $P = 0.0285$; LPS, $F_{(1, 16)} = 21.90$, $P = 0.0003$; Rescue, $F_{(1, 16)} = 2.736$, $P = 0.1176$, Fig. 7D]. These results suggested that microglia engulf synaptic material when challenged by LPS through a C1q-dependent manner.

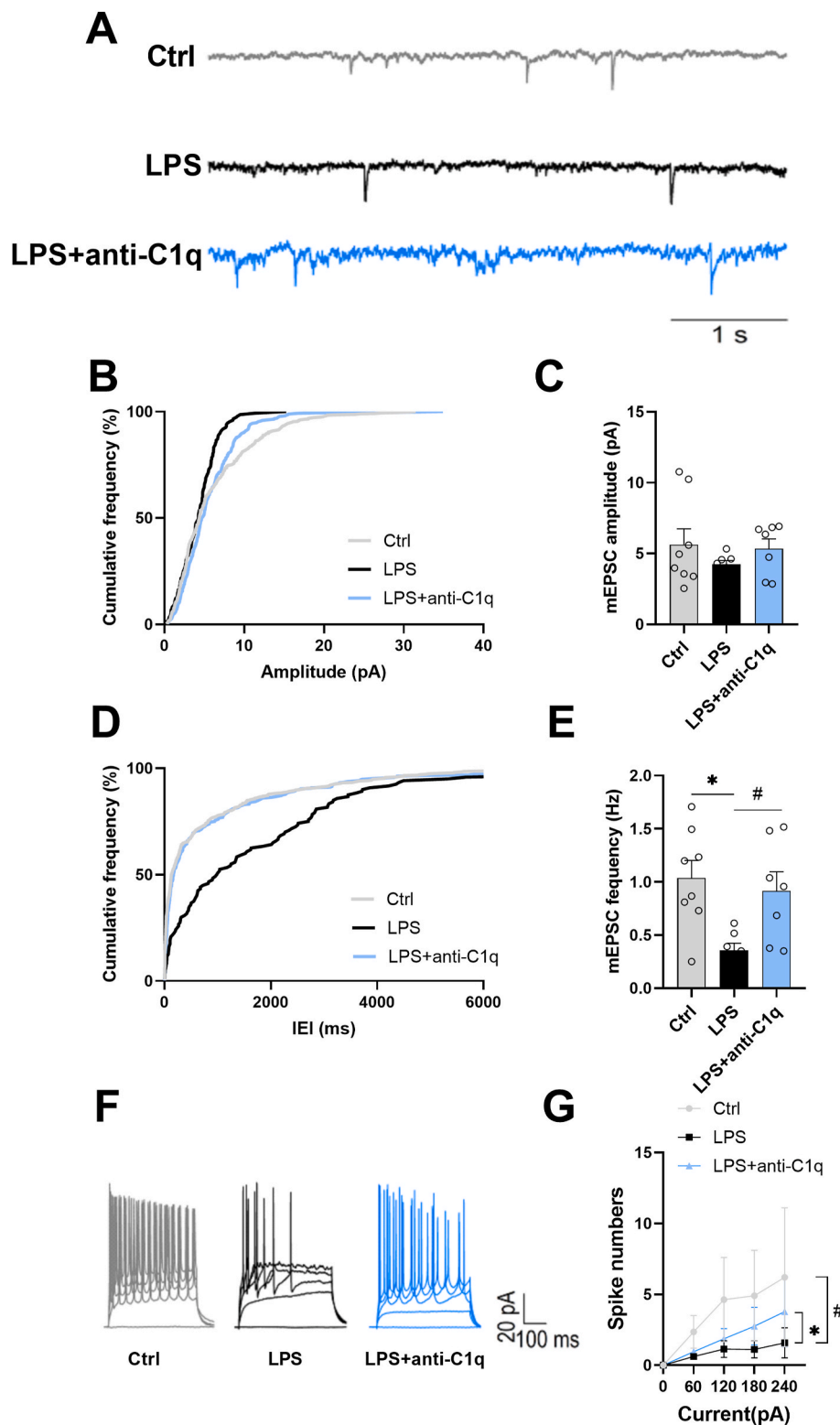


Fig. 4. C1q-neutralization antibody attenuated LPS-induced decrease in the membrane excitability and excitatory synaptic transmission in the hippocampal CA1. (A) Sample traces of mEPSCs were recorded from pyramidal neurons in CA1. (B–C) Quantification of mean amplitude of mEPSCs among three groups. (D–E) Quantification of mean frequency of mEPSC frequency among three groups. (F) Representative responses to currents (60 and 240 pA) recorded from pyramidal neurons. (G) Quantitative analysis of spike number among three groups. Data are presented as the mean $\pm$ SEM ($n = 7$ – 12), * $P < 0.05$, # $P < 0.05$.

3.8. C1q-neutralization antibody reversed LPS-induced abnormal theta oscillations in the hippocampal CA1

Memory acquisition and consolidation are associated with neuronal oscillations in the hippocampus. We therefore conducted *in vivo* electrophysiology to investigate the potential mechanism of LPS-induced cognitive impairments at the neuronal network level. Mice were placed to the Y maze to investigate the task-related LFP [Fig. 8A–B].

Compared with the control group, mice in the LPS group had a significantly lower power of theta oscillations in the Y maze. Moreover, C1q-neutralization antibody reversed the decreased power of theta oscillations [Interaction: $F_{(1, 23)} = 9.210$, $P = 0.0059$; LPS, $F_{(1, 23)} = 11.73$, $P = 0.0023$; Rescue, $F_{(1, 23)} = 1.602$, $P = 0.2183$, Fig. 8C], suggesting the potential role of C1q in modulating theta oscillation. The power of alpha, beta, gamma oscillations was not significantly affected by repeated LPS injections [Interaction: $F_{(1, 28)} = 0.4735$, $P = 0.4970$; LPS,

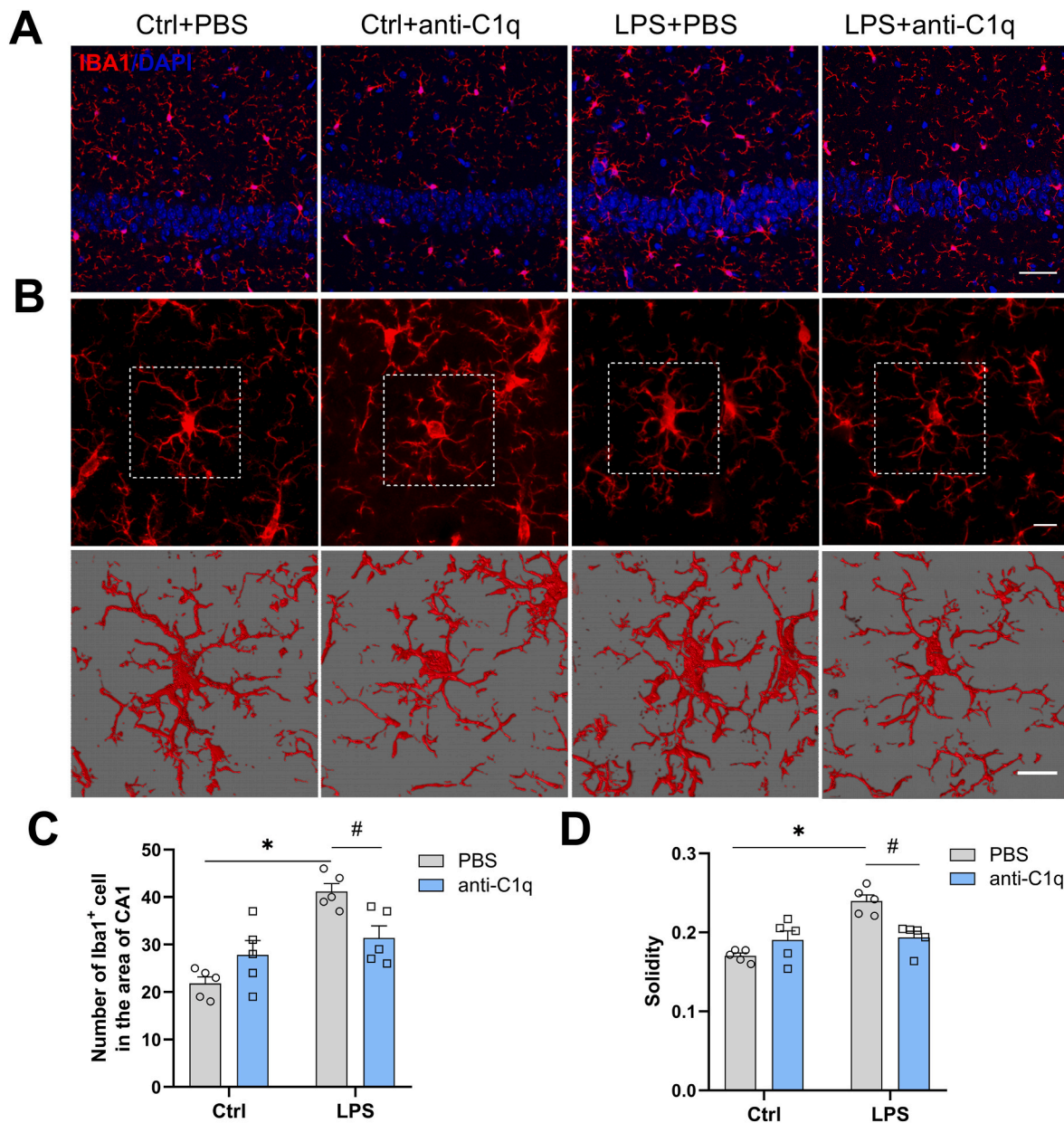


Fig. 5. C1q-neutralization antibody attenuated LPS-induced microglia activation in the hippocampal CA1. (A) Representative images of microglia in the CA1. Scale bar = 50 μ m. (B) Representative images of microglia morphology in the CA1. Scale bar = 10 μ m. (C) Quantitative analysis of the number of microglia in the CA1. (D) Quantitative analysis of microglia morphology in the CA1. Data are presented as the mean $\pm$ SEM (n = 5), * P < 0.05, # P < 0.05.

$F_{(1, 28)} = 4.910$, $P = 0.0350$; Rescue, $F_{(1, 28)} = 0.4118$, $P = 0.5263$, Fig. 8D; Interaction: $F_{(1, 28)} = 0.2419$, $P = 0.6267$; LPS, $F_{(1, 28)} = 0.3982$, $P = 0.5332$; Rescue, $F_{(1, 28)} = 0.0006826$, $P = 0.9793$, Fig. 8E; Interaction: $F_{(1, 28)} = 0.5533$, $P = 0.4632$; LPS, $F_{(1, 28)} = 2.436$, $P = 0.1298$; Rescue, $F_{(1, 28)} = 1.012$, $P = 0.3230$, Fig. 8F].

3.9. C1q-neutralization antibody attenuated LPS-induced decrease in number of c-fos⁺ cell and long-term potentiation impairment in the hippocampal CA1

We further investigated the effect of LPS on number of c-fos⁺ cell in the CA1. Our results showed that compared with control group, the number of c-fos⁺ cell decreased significantly in LPS group. Furthermore, C1q-neutralization antibody also reversed the decreased number of c-fos⁺ cell [Interaction: $F_{(1, 14)} = 12.13$, $P = 0.0037$; LPS, $F_{(1, 14)} = 21.45$, $P = 0.0004$; Rescue, $F_{(1, 14)} = 1.232$, $P = 0.2857$, Fig. 9A–B]. LTP is closely related to form learning and memory function. To determine

whether C1q is associated with LPS-induced LTP impairment, the fEPSP slope was measured [Fig. 9C–D]. LTP was impaired in the CA1 in the LPS group compared with the control group. In addition, the slope of LTP was decreased significantly in the LPS group as compared with those in control group, which was significantly rescued by C1q-neutralization antibody [Ctrl vs. LPS, $P = 0.0334$; LPS vs. LPS + anti-C1q, $P = 0.0403$, Fig. 9E].

4. Discussion

In the current study, we demonstrated that repeated LPS injections maintained an elevated microglial inflammatory phenotype and induced hippocampal synaptic loss, which ultimately resulted in cognitive impairments. More specifically, we provided evidence that complement C1q signaling mediated, at least in part, these neuropathological changes. Overall, our study showed that systemic inflammation finally reached to the brain to cause neuroinflammation with

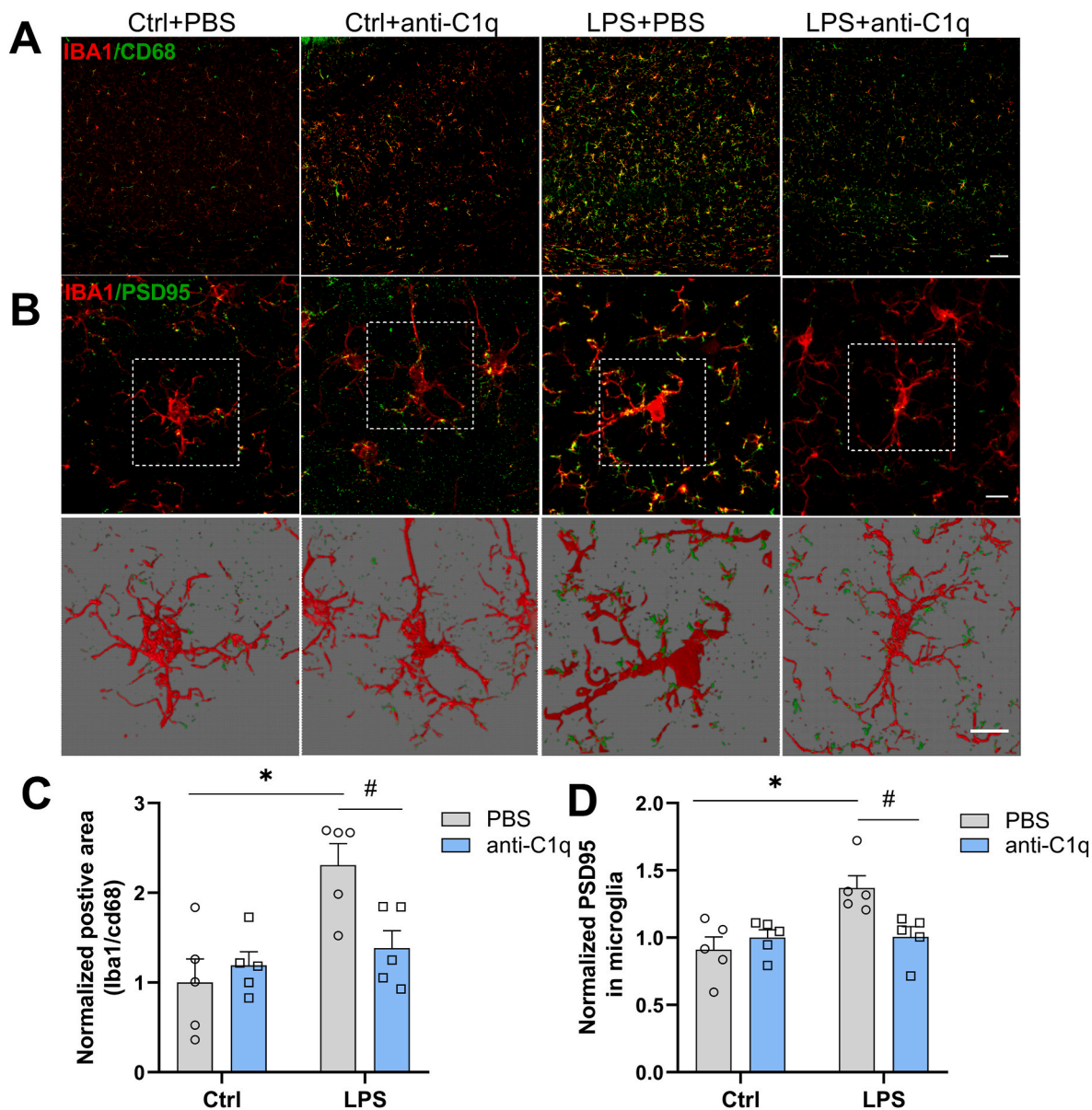


Fig. 6. C1q-neutralization antibody attenuated LPS-induced increase in microglia phagocytosis in the hippocampal CA1. (A) Representative images of CD68⁺ microglia cells in the CA1. Scale bar = 50 μ m. (B) Representative images of PSD95 puncta engulfed in microglia in the CA1. (C) Quantification of CD68⁺ microglia cells in the CA1. Scale bar = 10 μ m. (D) Quantification of the PSD95 puncta in microglia in the CA1. Data are presented as the mean $\pm$ SEM (n = 5 mice/group), * P < 0.05, # P < 0.05.

manifestation of microglia associated complement C1q modulated synaptic morphological and functional loss and ultimately cognitive impairment (Fig. 10).

Cognitive impairments frequently occurred following acute or chronic illness including infection, traumatic injury (e.g. orthopedic fracture) or surgery (Marra et al., 2018; Neuman et al., 2014). The hippocampus, one of the key brain regions for cognitive processing is particularly susceptible to sepsis or related immune priming insults induced neuroinflammation and subsequent hippocampal depending cognitive impairments (Smith et al., 2015; Shaker et al., 2021; Yirmiya and Goshen, 2011). Systemic inflammation is reported to worsen the CA3 synaptic changes in the ageing brain (Manabe et al., 2021). In addition, LPS-induced neuroinflammation resulted in synaptic defects, which was responsible for depressive-like behaviors (Li et al., 2021). Moreover, it has been also reported that the dysregulation of synaptic transmission in the basolateral amygdala projection neurons may account for neuroinflammation-induced anxiety-and depressive-like

behavior (Zheng et al., 2021). In our study, the repeated LPS injections reduced the number of dendritic spines and impaired synaptic plasticity, which were in line with previous studies that LPS-treated mice exhibited significant synaptic and neuronal damage in the hippocampus (Liu et al., 2018; Moraes et al., 2022; Kim et al., 2022). Overall, our data together with previous studies suggested that neuroinflammation-driven synaptic abnormality is a prominent pathogenic mechanism contributing to cognitive impairments. However, the underlying mechanisms by which systemic inflammation leads to synaptic and cognitive impairments remains to be elucidated.

Microglial cells are widely considered as the resident macrophages of the brain and function as active surveyors of the brain microenvironment in both healthy and neurodegenerative diseases (Liu et al., 2018). Under normal conditions, microglia display small cell bodies and multiple extending processes, which are responsible for the maintenance of brain homeostasis and contributing to the protection and remodeling of synapses (Paolicelli et al., 2011). Under disturbed or pathological

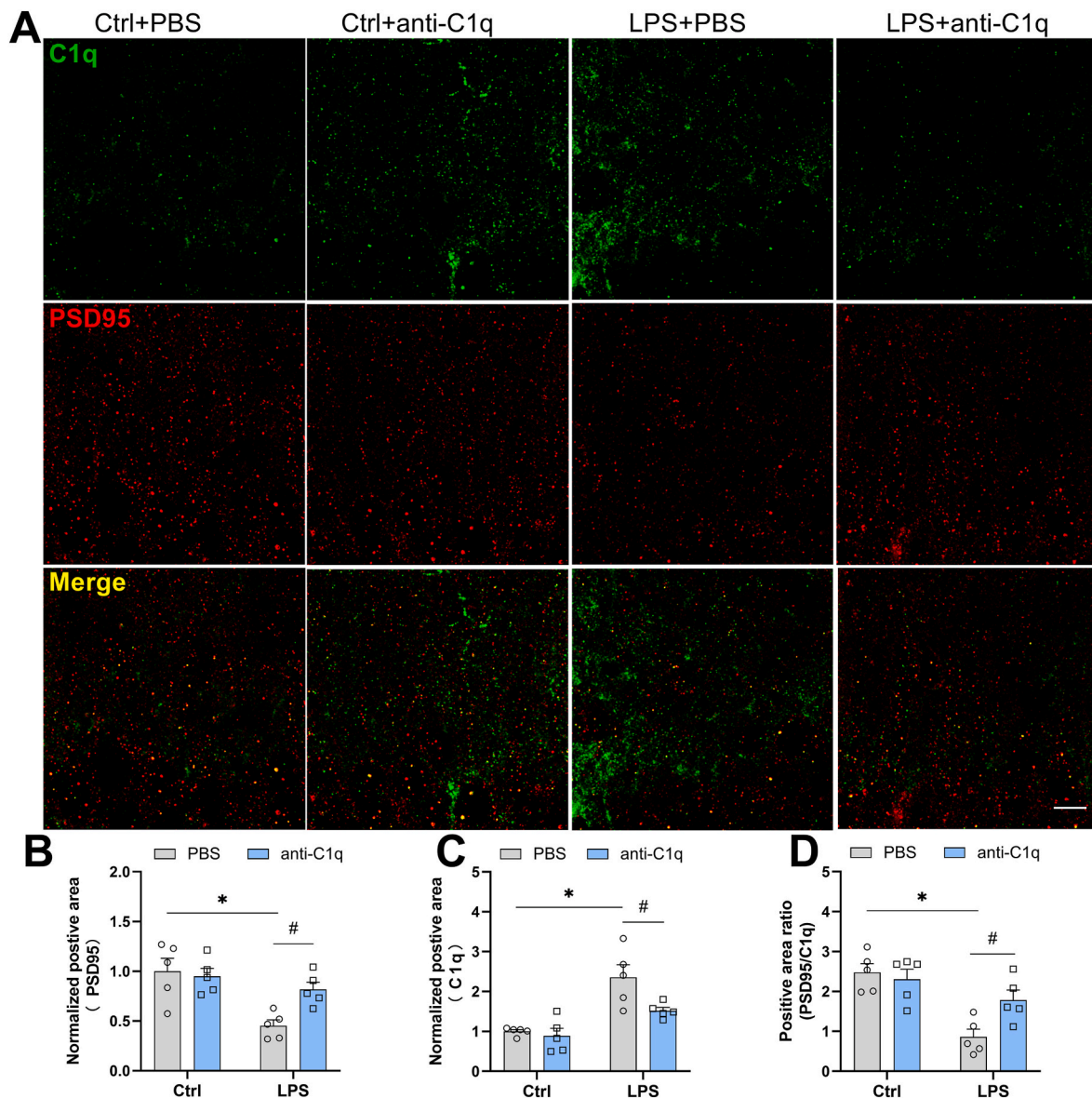


Fig. 7. C1q-neutralization antibody attenuated C1q-mediated microglial recognition and pruning of synapses in the hippocampal CA1. (A) Representative images of PSD95 puncta co-localized with C1q in the CA1. Scale bar = 10 μ m. (B) Quantitative analysis of positive area of PSD95 puncta in the CA1. (C) Quantitative analysis of positive area of C1q in the CA1. (D) Quantitative analysis of the ratio of positive area of PSD95 and C1q in the CA1. The ratio of positive area of PSD95 and C1q defined as the positive area of PSD95 divided by the positive area of C1q. Data are presented as the mean $\pm$ SEM ($n = 5$ mice/group), * $P < 0.05$, # $P < 0.05$.

conditions, microglia became an activated phenotype, morphologically characterized by a reduction in branch number accompanied by an increase in cell soma volume (Guedes et al., 2022; Liu et al., 2018). Microglial activation was reported to have enormous detrimental effects on neurons (Hong et al., 2016). In our study, the activated or amoeboid microglia were phagocytic and displayed enlarged cell somas with one or less extending process. Importantly, LPS-exposed mice showed higher intensity of CD68-positive phagocytic microglia, suggesting that reduced synaptic puncta may be due to increased microglial engulfment. In particular, repeated LPS injections were found to be mainly phagocyte excitatory synapse, although we cannot exclude its effects on inhibitory or excitatory synapses which were found to be engulfed by microglia previously (Morales et al., 2022; Chen et al., 2014). To further investigate the mechanism underlying microglial synaptic engulfment, we performed transcriptomic and proteomic analysis. In our study, we identified genes associated with microglial-mediated phagocytosis, especially for the complement C1q. The complement system is a major constituent of the innate immune system, mainly involved in defence

against pathogens (Hong et al., 2016; Vasek et al., 2016). Complement factors are significantly up-regulated in the aged brain and in several CNS neurodegenerative diseases, leading to neuronal damage, synaptic loss, and cognitive impairments (Hong et al., 2016; Vasek et al., 2016; Alawieh et al., 2020; Bourel et al., 2021; Xiong et al., 2018). For instance, excess C1qa is deposited on synapses before the appearance of plaques in the hippocampus of several AD mouse models (Hong et al., 2016). On the other hand, complement C3 and C3aR are found to mediate presynaptic terminal loss in the hippocampus of mice and induce spatial learning defects during recovery from West Nile neuro-invasive disease (Vasek et al., 2016). Consistently, microglia and C1q complement activation was found in hippocampal autopsy tissue of patients with sepsis, while specific C1q-blocking antibody prevented microglial engulfment of C1q-tagged synapses and improved neuro-cognitive outcome in sepsis mice (Chung et al., 2023). In addition, ablation of C3a receptor 1 rescued synapse loss in tauopathy and AD models (Litvinchuk et al., 2018). Thus, the complement pathway influence a wide range of phenotypes associated with neurodegeneration

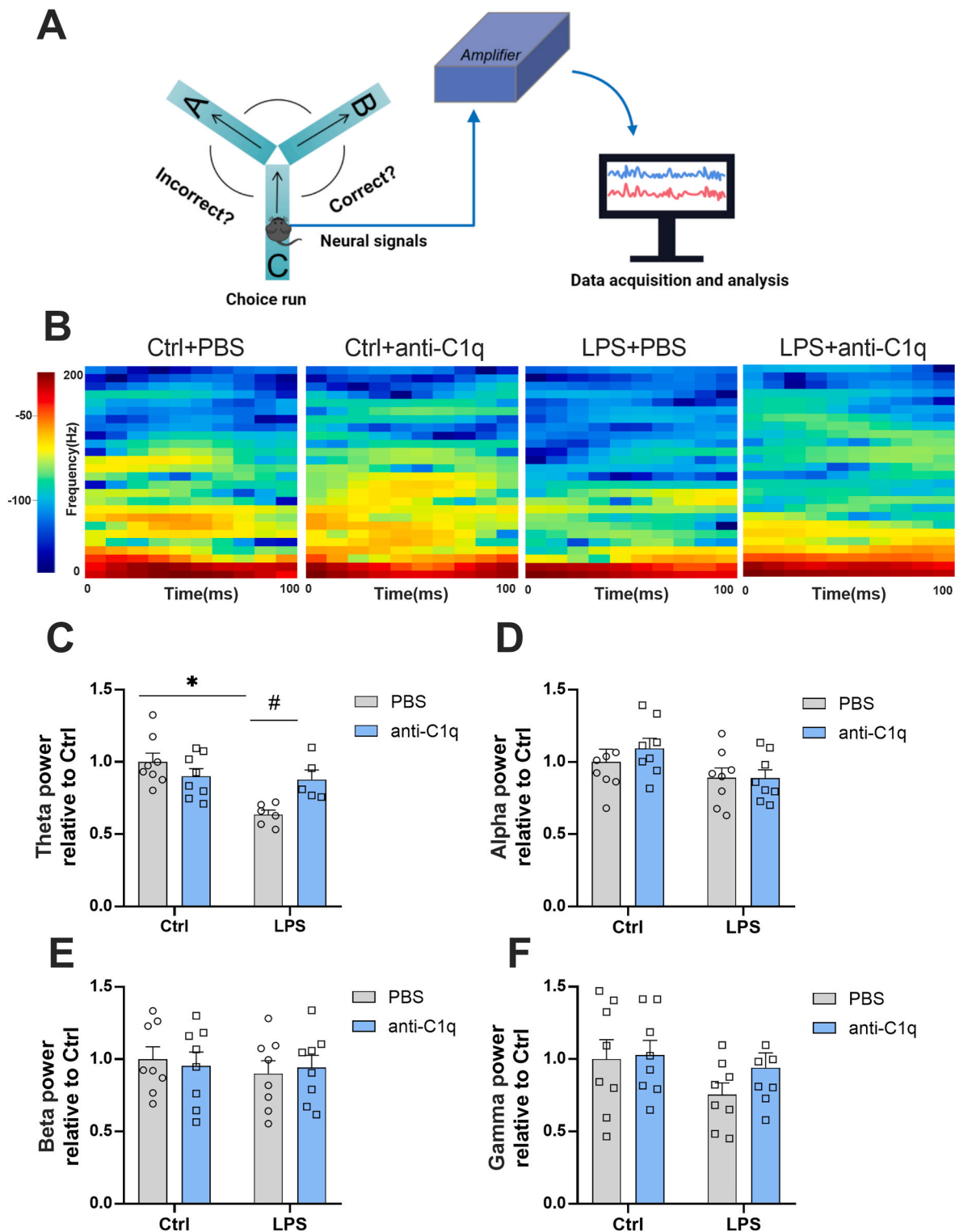


Fig. 8. C1q-neutralization antibody reversed LPS-induced abnormal neuronal oscillations in the hippocampal CA1. (A) Schematic representation of the working memory task in Y-maze. (B) Time-frequency diagrams of LFP. (C) The power of theta oscillation in Y maze task. (D) The power of alpha oscillation in Y maze task. (E) The power of beta oscillation in Y maze task. (F) The power of gamma oscillation in Y maze task. Data are shown as mean $\pm$ SEM ($n = 5-8$ mice/group), $*P < 0.05$, $^{\#}P < 0.05$.

by modulating glial responses and contributing to neuronal injury and synapse loss. Our study further revealed that neutralization of c1q signaling significantly prevented these synaptic loss and cognitive impairments in mice receiving repeated LPS injections. Our data demonstrated that complement c1q has been implicated in microglia-mediated

synapse loss. However, the major cells that contribute to production of proteins of the complement system are microglia and, to a lesser extent, astrocytes (Hong et al., 2016). Thus, we cannot completely rule out the contribution of astrocytes or other cell types to this process.

Furthermore, although we excluded the potential crystallizable

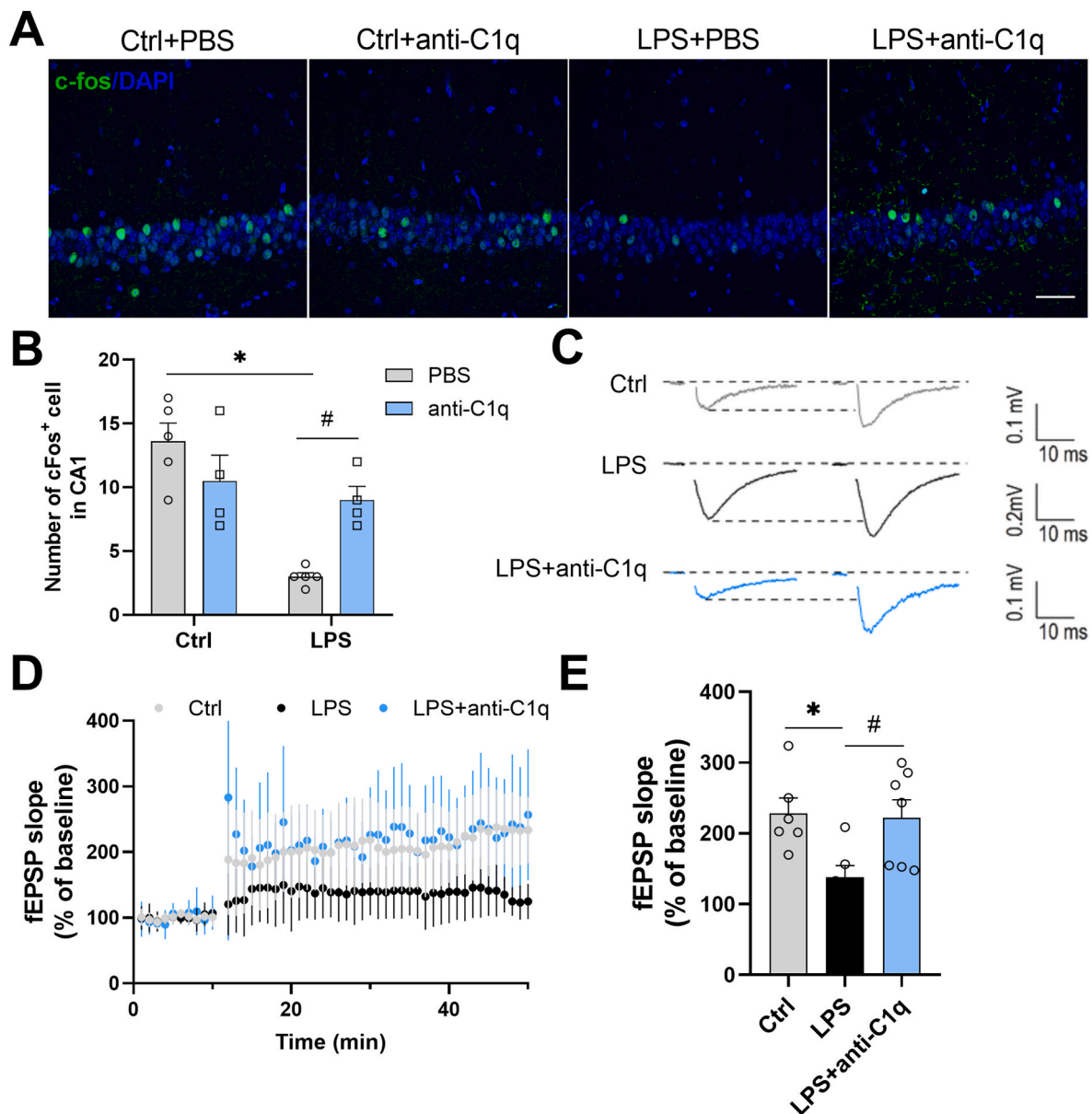


Fig. 9. C1q-neutralization antibody attenuated LPS-induced decrease in number of c-fos⁺ cell and long-term potentiation impairment in the hippocampal CA1. (A) Representative images of c-fos⁺ cell in the CA1. Scale bar = 10 μ m. (B) Quantitative analysis of the number of c-fos⁺ cell in the CA1. (C) Effects of the C1q-neutralization antibody on LTP of the hippocampal neurons. (D-E) Quantitative analysis of fEPSP slope in the CA1. Data are presented as the mean $\pm$ SEM (n = 5–6 mice/group), * P < 0.05, # P < 0.05.

fragment (Fc) domain-mediated functions of the c1q-neutralization antibody in our study (Fig. S4), a well-established Annexon c1q antibody with silenced Fc part may be a better intervention. In addition, microglia appear to play a key homeostatic role in regulating neuronal activity by synaptic remodeling in CNS (Schafer et al., 2012). It has been demonstrated that microglia depletion increases neural activities of excitatory neurons (Liu et al., 2021). Our study showed that LPS induced excitatory synapse loss, excitatory synaptic transmission and neuronal firing, which was prevented by c1q-neutralization antibody. These results suggested the key role of microglia in mediating synapse pruning and neuronal activity regulation. Consistently, early life inflammation induced excessive microglial engulfment and caused the impaired neuronal activity in the anterior cingulate cortex (Cao et al., 2021). These results indicate that LPS-induced neuroinflammation increases the capacity of microglia to engulf excitatory synapse, triggers abnormal neuronal transmission and ultimately leads to decreased neuronal firing.

Accumulating evidence has suggested that memory encoding,

consolidation, and retrieval require temporal binding of pyramidal cell assemblies for the generation of rhythmic oscillations (Liu et al., 2021). There are multiple different classes of network oscillations, which span a wide range of frequencies from 0.05 to 600 Hz. The major patterns of rhythmic oscillations in cortical networks include theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–100 Hz) frequency oscillations, which synchronize the action potential generation of principal cells with great precision (Gordon, 2011). In particular, theta oscillations play a key role in the control of working memory (Riddle et al., 2020). Indeed, theta frequency transcranial alternating current stimulation have been proved to improve working memory performance (Alekseichuk et al., 2016). In our study, we used LFP recordings to verify network activity states and showed that repeated LPS injections impaired theta oscillation but had no effect on other types of network oscillations. Importantly, this impaired theta oscillation was prevented by c1q neutralization. Thus, disrupted theta rhythm may represent the neuroinflammation mediated alterations of cognitive function *per se*. In

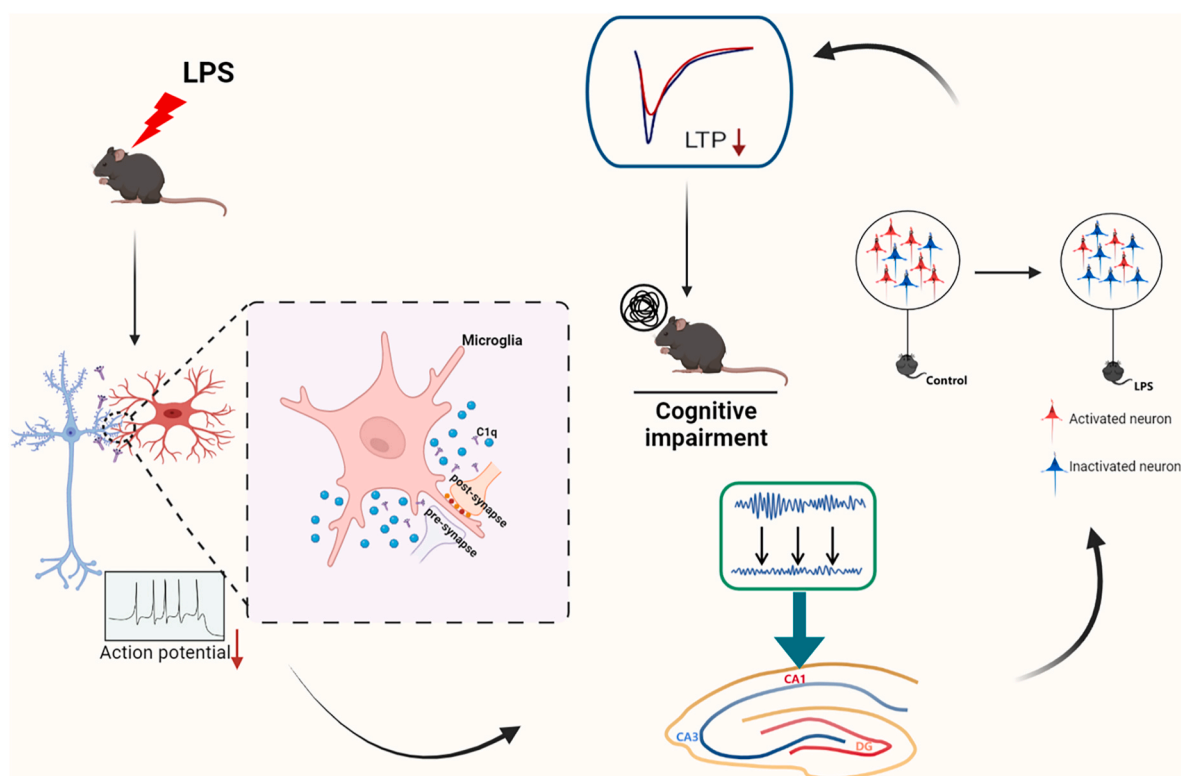


Fig. 10. Working model of this study. Systemic repeated LPS injections induced an elevated microglial inflammatory phenotype and induced hippocampal synaptic loss, ultimately contributing to abnormal neuronal oscillations and cognitive impairments, which is mediated by complement C1q signaling.

previous studies, altered network oscillations are observed in different sepsis models (Ji et al., 2020b; Yin et al., 2023). For instance, repeated LPS (1 mg/kg) injections for three times induced decreased theta, beta, and gamma oscillations in the medial prefrontal cortex when animal performing the novel objective recognition test (Ji et al., 2020b). In an animal model of sepsis induced by cecal ligation and puncture, only decreased theta oscillation was observed in the hippocampus (Yin et al., 2023). Thus, changes in network oscillations may depend on different experimental conditions, including different animal models used, different time for measurement, as well as different brain regions investigated.

Synaptic plasticity has long been considered a cellular mechanism of memory formation. Long-lasting forms of synaptic plasticity such as LTP are fundamental cellular mechanisms underlying learning and memory (Lisman et al., 2018). Many forms of synaptic plasticity induced at different synaptic connections during hippocampal oscillations may contribute to the formation of stable neuronal ensembles (Çalışkan and Stork, 2018). Most studies investigated the impact of oscillations on synaptic plasticity and suggested that oscillatory status could influence synaptic plasticity and cognition (He et al., 2021; Paterno et al., 2021). In the brain, neuronal gene expression is dynamically changed in response to neuronal activity. In particular, the expression of immediate-early genes such as *c-fos* is rapidly and selectively up-regulated in subsets of neurons in specific brain regions associated with learning and memory (Pettit et al., 2022). In the present study, repeated LPS injections reduced *c-fos* positive neuronal numbers in the CA1 area of the hippocampus after behavioral test, which was accompanied by significantly impaired LTP, while *c1q* neutralization rescued the reduced *c-fos* positive neuronal numbers, impaired LTP and cognitive impairments. These results suggested that disturbed disrupted theta oscillation and impaired LTP may be the mechanisms for the cognitive impairments found in this mouse model of neuroinflammation.

In summary, our data, for the first time, provided evidence that activated microglia in the hippocampus following systemic

inflammation and finally neuroinflammation by LPS initiated complement *c1q* cascade in inducing synaptic loss and theta oscillation impairment and consequent cognitive impairments. Anti-*c1q* therapy attenuated and/or prevented all these changes induced by LPS, indicating that complement *c1q* would be a therapeutic target for developing therapy to improve cognitive impairments induced by neuroinflammation. However, given the nature of an animal study reported here, further study to verify these findings are required. On the other hand, neuroinflammation was considered to be a common pathological condition in all neurodegeneration diseases. Whether the role of complement cascade found in the current study reported here could apply to those conditions remains elusive and warrants further study.

Ethics approval

The Ethics Committee of Animal Care and Use Committee of Nanjing Medical University, Jiangsu, China, approved all animal experiments described in this study.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Code availability

Not applicable.

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Declaration of competing interest

The authors declared that they have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix ASupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.neuropharm.2023.109646>.

Abbreviations

AD	Alzheimer's disease
LPS	lipopolysaccharide;
CNS	central nervous system
PSD95	post-synaptic density protein 95
PFA	paraformaldehyde
KEGG	Kyoto Encyclopedia of Genes and Genomes
LFPs	Local field potentials

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